Mathematical Modeling through Inductive Reasoning with LLMs

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Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in mathematical reasoning and data analysis through natural language interaction. Yet the extent to which LLMs can perform mathematical modeling via inductive reasoning, a specific form of mathematical reasoning that involves inferring functional relationships solely from numerical data, remains largely underexplored. We study this capability by evaluating whether LLMs can analyze numerical data and extract symbolic expressions capturing relationships between independent and dependent variables. To isolate inductive reasoning performance from confounding factors, we use a controlled, noise-free dataset that facilitates clear pattern matching and explicit cause-effect attribution. Our analysis focuses on in-context learning through structured reasoning strategies, such as backtracking and search-based prompting, which expose models to intermediate hypotheses and revision steps. These structured reasoning strategies improve accuracy by up to 37% on the full task compared to chain-of-thought prompting. Additionally, our framework decomposes the problem into subtasks including cause–effect identification and symbolic expression formation. Subtask analysis reveals both strengths and limitations: while models achieved 92% accuracy in cause–effect identification with ordered data, performance dropped below 45% when datapoints were permuted, underscoring challenges with invariance to input ordering. Our findings shed light on the extent to which LLMs can engage in interpretable, data-driven mathematical reasoning, offering a benchmark for inductive reasoning in mathematical contexts and insight into LLMs' limitations and their broader role in automated modeling and numerical analysis.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in mathematical reasoning and data analysis through natural language interaction. Yet the extent to which they can perform mathematical modeling via inductive reasoning, i.e. inferring functional relationships solely from numerical data, remains largely underexplored. Existing benchmarks such as ACRE and MiniSCAN (Fig. 1) test inductive reasoning on symbolic or visual data; our novel benchmark is the first to do so in a purely numerical setting. Unlike conventional math word problems, our task removes linguistic cues and priors: the model is presented with purely numerical input-output pairs, requiring it to infer the governing equation from data alone (Fig. 2). This isolates inductive reasoning from memorization or linguistic shortcuts.

This problem setting is important for understanding current LLM capabilities. After all, despite excelling in complex tasks such as solving Olympiad-level problems [9], generating code [7], and performing multi-hop reasoning [17], LLMs often fail on basic arithmetic and simple data-driven

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inference [16], suggesting that their reasoning may rely more on pattern recognition and memorization than on conceptual understanding. Hence, we design controlled datasets that avoid noise and external priors, enabling clear evaluation of inductive reasoning. To probe model behavior more precisely, we also decompose the task into subtasks: (i) identifying cause–effect relationships between variables and outputs, and (ii) composing these into symbolic equations.

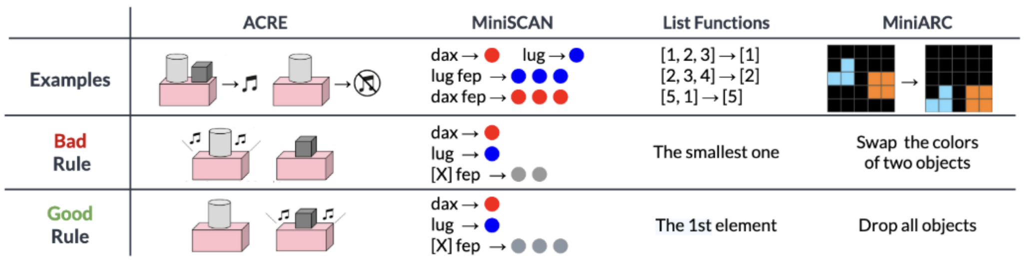

Figure 1: Inductive reasoning tasks across four benchmarks (ACRE, MiniSCAN, List Functions, MiniARC), showing examples, a correct rule, and an incorrect generalization for each, from [11]. This highlights how models infer patterns from examples—analogous to our task, but our task involves strictly numerical data without any priors.

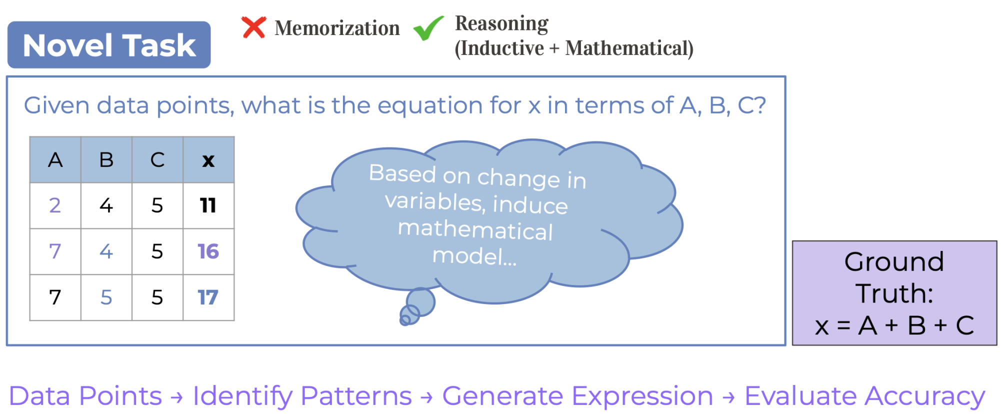

Figure 2: To evaluate LLMs' ability to perform numerical mathematical modeling through inductive reasoning (not memorization), we introduce a novel benchmark: instead of word problems, the model is given only raw numerical input-output pairs in table form. For example, increasing A by 5 increases x by 5, and increasing B by 1 increases x by 1—indicating linear contributions. These example datapoints show how variable perturbations reveal cause-effect structure. The model must detect such patterns and infer the ground truth equation (e.g., $x = A + B + C$). This tests its ability to generalize from examples, identify variable relationships, perform arithmetic, and express symbolic equations.

We center our study around two guiding research questions:

- **RQ1: To what extent can LLMs perform mathematical modeling via inductive reasoning?** We evaluate whether LLMs can extract symbolic equations from structured data and analyze their strengths and limitations across subtasks.
- **RQ2: How can structured reasoning strategies improve inductive reasoning performance?** We investigate backtracking and search-based prompting, which expose models to intermediate hypotheses and revision steps, improving accuracy by up to 37% over chain-of-thought prompting.

By addressing these questions, we contribute (i) a benchmark framework for evaluating LLMs' inductive reasoning abilities in numerical contexts and (ii) insights into strategies that enhance interpretability and accuracy of inductive reasoning with LLMs.

Our experiments reveal three central findings. First, small LLMs (<32 billion parameters) perform poorly on this mathematical modeling task, often defaulting to simplistic linear predictions even for nonlinear ground truth equations. Second, larger models demonstrate some ability to recover ground-truth equations but remain fragile: they perform well on structured datapoints yet collapse when input order is permuted, showing reliance on sequential heuristics rather than robust reasoning. Third, structured reasoning strategies, such as search-based prompting and backtracking, substantially improve performance, boosting accuracy by up to 37% over standard chain-of-thought prompting. Overall, our findings shed light on the extent to which LLMs comprehend mathematics and inform future directions for fine-tuning and analyzing inductive reasoning mechanisms (e.g., attention weights).

2 Related Work

Inductive reasoning benchmarks. LLMs demonstrate inductive reasoning via in-context learning, generalizing from examples to infer rules [2]. Existing benchmarks such as ACRE, MiniSCAN, and MiniARC evaluate this ability on symbolic, string, or visual data (Fig. 1) [11], but not on purely numerical tasks. Our work extends this line by introducing controlled numerical datasets, deliberately eliminating linguistic cues and priors to isolate inductive reasoning.

Numerical reasoning with LLMs. Prior work evaluates LLMs on noisy, multimodal real-world data [8], whereas our controlled setup targets deterministic relationships with known ground truth. This design allows systematic variation in task complexity (e.g., number of variables, linear vs. non-linear equations, coefficient scaling) to probe reasoning depth. While LLMs can solve Olympiad-level problems [9], generate code [7], and perform multi-hop reasoning [17], they often fail at basic arithmetic [16]. They are strong at proposing hypotheses but weak at verification, especially for factual lookups or calculations [13, 5]. To mitigate this, prior studies integrate external tools such as calculators [13], Python interpreters [5], or optimizers within tool-calling agents. Other approaches emphasize structured reasoning strategies, including search, backtracking, and hypothesis refinement [11, 18, 4], and we build on these methods in a numerical setting.

Numerical puzzles such as Countdown and the Game of 24 also probe numerical reasoning. These tasks require combinatorial search over arithmetic operations to reach a target value, with multiple valid solutions in Countdown and exactly one in the Game of 24. Performance is measured by proximity to the target (for Countdown) or exact correctness (for Game of 24) [4, 18]. While they test LLMs' search capabilities, they do not evaluate inductive reasoning. In contrast, our task requires discovering the symbolic expression that generated the data, thereby directly targeting rule inference.

Symbolic regression. Our task also parallels symbolic regression, where the goal is to recover mathematical models from data [14, 6, 10]. However, prior works typically use noisy real-world datasets, making exact rule recovery impossible without external optimizers. In those settings, LLMs may identify the functional form but rely on external tools to fit constants. By contrast, our controlled setup (Fig. 2) permits exact recovery without optimizers, allowing us to probe inductive reasoning directly.

Applications in education and science. Finally, our work aligns with applications in the ML4ED lab [3, 12, 15], where agents interact with simulations such as PharmaSim and PhET. These agents often struggle with numerical reasoning due to poor data collection. Improved inductive reasoning could enhance student simulators, curriculum validation, and scientific modeling across domains such as glucose dynamics, climate modeling, and biology [8, 1].

3 Methods

3.1 Dataset Construction

We design four purely numerical synthetic datasets, each with 27 symbolic equations progressively increasing in difficulty (linear $\rightarrow$ nonlinear $\rightarrow$ larger coefficients), to isolate LLMs' inductive reasoning from confounding priors such as natural language cues. All datasets use three independent variables (A, B, C) and one dependent variable x. For each equation, we generate 19 datapoints where consecutive datapoints have differing values in only one independent variable, such as the example shown in Fig. 2. In table 1 and below, we further describe the construction of the datasets.

Table 1: Datasets Overview

Dataset	**Description**
`linear_eqs`	Linear, with coefficients
`eqs_1`	Linear & nonlinear, no coefficients
`eqs_2`	Linear & nonlinear, with coefficients
`eqs_3`	Linear & nonlinear, with larger coefficients

3.1.1 Datapoint Construction

Once the ground-truth equation is fixed, datapoints are generated in the same way for all datasets. We first sample initial values for (A, B, C) from $[-10, 10]$ and compute the corresponding x from the ground-truth equation. For each subsequent datapoint, exactly one variable is perturbed by a value drawn from $[-5, 5]$, while the other two remain fixed at their previous values. This procedure continues until 19 datapoints are obtained. By construction, this setup provides a clear inductive signal: models can isolate the contribution of each variable by observing successive changes across consecutive datapoints.

3.1.2 Linear Equations (`linear_eqs`)

This dataset contains 27 equations restricted to linear combinations of the three variables (A, B, C). Each ground-truth equation has the form $x = \alpha A + \beta B + \gamma C$ where the coefficients α, β, γ are sampled uniformly from the interval $[-5, 5]$.

3.1.3 Equations 1 (`eqs_1`)

This dataset consists of 27 linear and non-linear expressions systematically generated from the three variables (A, B, C). We enumerate all possible combinations of the four basic arithmetic operations $\{+, -, \times, \div\}$, including bracketed groupings to capture different operator precedences, and randomly select 27 ground-truth equations. In contrast to the linear dataset, all coefficients are fixed to 1, ensuring that variation arises solely from the structure of the expression rather than from parameter values.

3.1.4 Equations 2 and Equations 3 (`eqs_2` and `eqs_3`)

These datasets have 54 equations total (27 each) and build directly on `eqs_1` by introducing integer coefficients, sampled from $[-5, 5]$, applied to both variables and sub-expressions. This extension increases algebraic complexity by combining operator composition with scalar multipliers.

3.2 Experiment Pipeline

To evaluate LLMs on the inductive reasoning benchmark introduced in Fig. 2, we employ a standardized experimental pipeline. Each dataset consists of datapoints $D = \{(A_i, B_i, C_i, x_i)\}_{i=1}^{n}$, where x is generated by a ground-truth function f such that $x = f(A, B, C)$. The maximum number of datapoints provided is $n = 19$. Models receive a prompt P containing task instructions, the datapoints, and—when applicable—in-context learning (ICL) examples. Given this input, a model L outputs a predicted equation $\hat{f} = L(X, P)$. Predictions are evaluated for *symbolic accuracy*, defined as an exact match between $\hat{f}$ and f, up to commutativity and associativity. We also manually analyze the nature of incorrect predictions to observe possible patterns and biases in reasoning.

This setup enables systematic comparison across model sizes, equation complexities, and prompting strategies. We test direct prompting, chain-of-thought (CoT) prompting, and ICL with structured reasoning traces such as search and backtracking. This controlled design allows us to evaluate adaptation strategies, expose biases, and diagnose reasoning bottlenecks without modifying model weights.

We evaluate four LLMs spanning a range of parameter sizes: GPT-4o-mini, GPT-5 mini, DeepSeek-32B, and Qwen-14B. In this work, models with fewer than 32 billion parameters (GPT-4o-mini and Qwen-14B) are treated as "smaller" models, while GPT-5 mini and DeepSeek-32B are considered "larger." To control for datapoint effects, we focus on 10 and 13 datapoints, which consistently yield the highest symbolic accuracy across models before performance plateaus (Fig. 3).

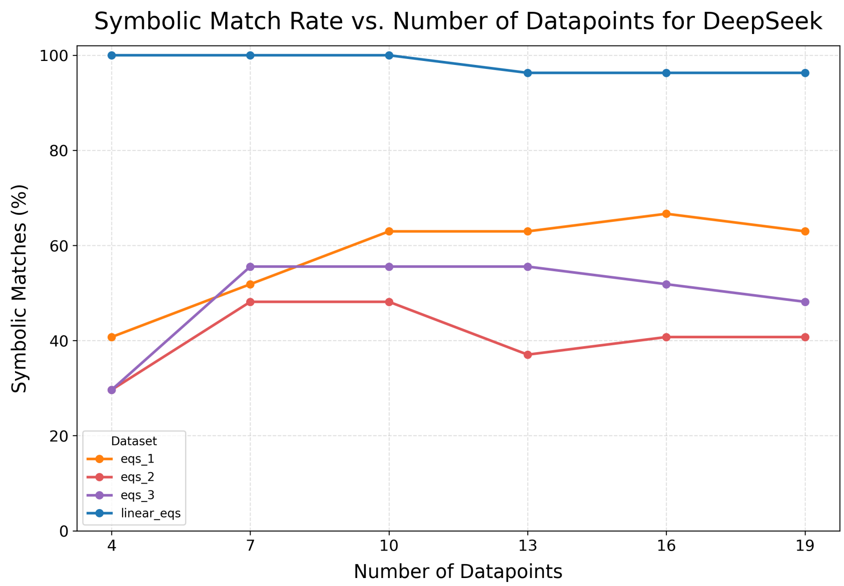

Figure 3: Symbolic match rate as a function of datapoint count for DeepSeek (similar results were observed for the other three models considered in this work). Accuracy increases with additional datapoints but plateaus at 10 and 13, showing that 10 or 13 datapoints provide sufficient amount of data for evaluation of model performance without redundancy.

3.2.1 Baseline Prompting Strategies

Direct Prompting: Models receive datapoints and are asked to produce an equation without being given an explicit reasoning strategies in a zero-shot setting.

Chain-of-thought (CoT): Models receive datapoints and are encouraged to employ step-by-step reasoning before generating a final hypothesis.

3.2.2 In-Context Learning (ICL) Strategies

To address failure patterns observed in baseline experiments (linearity bias, cascading errors from poor initial hypotheses), we introduce hand constructed high quality ICL examples that illustrate how to overcome important reasoning bottlenecks that were found in baseline prompting strategies through strategies such as backtracking when revision of an initial hypothesis is necessary and testing final hypotheses on all datapoints to verify accuracy, and backtracking in case of failure. We provide 10 ICL examples for the full task, 5 of which illustrate linear reasoning (optimal) and the other 5 illustrate nonlinear reasoning (backtracking). These are based on Stream of Search (SoS) [4] and Backtracking strategies, described in Fig. 4 and as follows.

SoS traces illustrate iterative hypothesis generation, evaluation against datapoints, and refinement.

Backtracking traces explicitly encode revision steps when an initial guess fails, enforcing goal-checking behavior. Without revision steps, baseline results show that models often commit to incorrect linear hypotheses; backtracking enforces systematic revision when predictions fail.

Models are tested with zero-shot (no ICL examples) and few-shot (varying types of ICL examples) throughout the experiments.

```
- node: string index of the current node (e.g., "0", "0.0", "0.0.1", etc.)
  current_state:
    hypothesis: the current symbolic expression being tested (or null at the
                start)
    evidence: a list of verbal conclusions from earlier steps
    reasoning: a list of logical justifications for each step taken
  action: one of:
    - "<<EXPAND>>" (to add a new reasoning step)
    - "<<BACKTRACK>>" (to abandon the current path and return to a previous node)
    - "<<EXPAND_AFTER_BACKTRACK>>" (to try a new structure after backtracking)
    - "<<GOAL_CHECK>>" (to evaluate the final hypothesis on all datapoints)
  operation: what you're doing (e.g., "Compare dp1 and dp2 to isolate B")
  datapoints: the relevant datapoints being analyzed
  observation: what you observed from the changes (e.g. change in x with
               respect to a variable)
  updated_state: updated hypothesis, evidence, and reasoning after this step
    hypothesis: updated symbolic form for x (e.g., "x = A + 2*B + ___")
    evidence: cumulative evidence supporting this hypothesis
    reasoning: cumulative logical reasoning trace so far
  result: Whether the hypothesis fits all datapoints
  final_hypothesis: The final expression for x in terms of A, B, C, exactly
                    matching all datapoints
```

Figure 4: Structured reasoning trace schema. The figure illustrates the schema used for representing inductive reasoning in ICL demonstrations. Each node corresponds to an intermediate reasoning step, where the model maintains a symbolic hypothesis, supporting evidence, and justification. At every inductive step (i.e., node), the model executes an action such as expand (add a new reasoning step), backtrack (go to parent node), expand-after-backtrack (try a new structure/node after revision), or goal-check (evaluate the final hypothesis against all datapoints). This explicit structure encourages models to reason through intermediate hypotheses, revise incorrect assumptions, and search systematically over candidate expressions, culminating in the final hypothesis that matches all datapoints.

3.3 Subtask Decomposition

To detect reasoning bottlenecks, we decompose the fulltask into 2 subtasks:

1. **Cause-effect Identification**: Determine the effect of changing a single independent variable on the dependent variable x. Example: $+4A \rightarrow +4x$.
2. **Mathematical Expression Formation**: Combine variable cause-effects identified in Subtask 1 into a final symbolic rule (i.e., the mathematical expression). Example: $+4A \rightarrow +4x$, $+B \rightarrow +x$, $+3C \rightarrow +3x \Rightarrow x = A + B + C$

This decomposition distinguishes between failures in variable attribution and failures in compositional equation building. We provide 5 ICL examples for each subtask using equations from datasets of varying difficulty.

4 Results

4.1 Baseline Performance

We first evaluated models on the full modeling task under direct and chain-of-thought (CoT) prompting. Results reveal strong baseline limitations. The smaller models (<32B; GPT-4o mini and Qwen-14B) consistently failed to produce correct symbolic equations, often defaulting to linear hypotheses even when the ground truth equations is nonlinear. This linearity bias appeared in over 90% of Qwen's initial hypotheses and typically led to cascading errors, as the model rarely revised incorrect assumptions.

By contrast, larger models (DeepSeek-32B and GPT-5 mini) achieved near-perfect accuracy on the linear dataset and non-trivial accuracy on the nonlinear sets, but tended to rely on long reasoning traces (often thousands of output tokens). CoT helps the larger models, with the benefit increasing as equation complexity grows (e.g., `eqs_1`$\rightarrow$`eqs_3`). Small models have below 25% symbolic match rates across all datasets even with CoT, indicating a limitation of performance based on model size for baseline experiments. Overall accuracy across all conditions is fairly low (mean $\approx$ 37%), and Fig. 5 summarizes these trends: CoT boosts capable models, while small models remain ineffective regardless of dataset complexity.

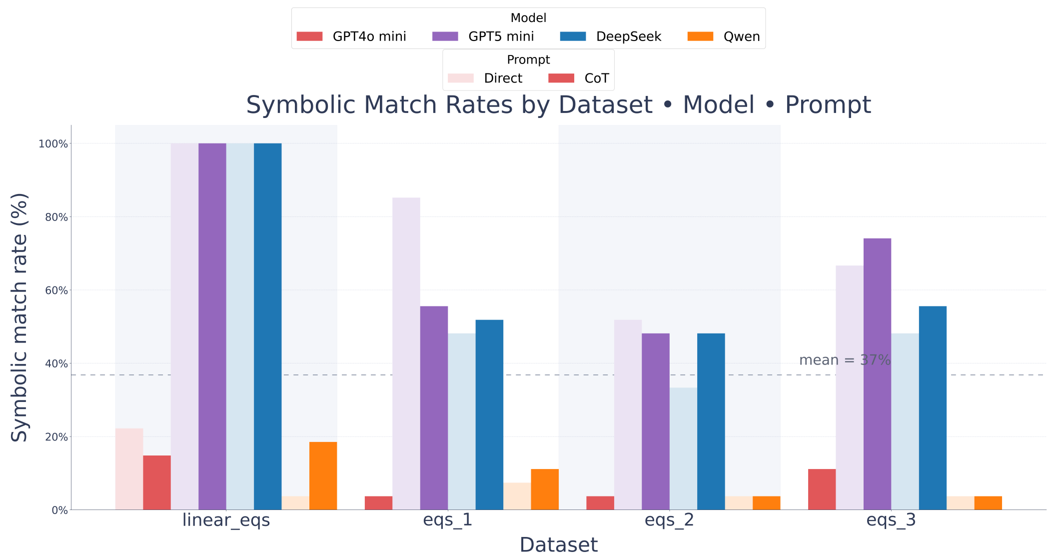

Figure 5: Clustered bar plot showing symbolic accuracy across datasets, models, and baseline prompting styles. Larger models benefit from CoT prompting as the equation complexity increases, underscoring the role of structured reasoning scaffolds. However, smaller models ($<$ 32B) consistently have low symbolic match rates ($<$ 25%) regardless of equation complexity, revealing performance limitations for smaller models in baseline experiments.

4.2 Performance under Structured Reasoning Strategies

We explore whether structured reasoning strategies can mitigate baseline prompting strategy limitations. Incorporating in-context learning (ICL) with reasoning traces, specifically Stream of Search and Backtracking demonstrations (Section 3), yielded substantial improvements. Providing ICL examples of how to hypothesize, test, and revise equations raised symbolic accuracy by as much as 37% compared to direct or CoT prompting as illustrated in Fig. 6.

The gains were most pronounced for larger models. GPT-5 mini exceeded 80% symbolic match rate for all datasets, while Qwen and GPT-4o mini remained below 42%, even with ICL. As shown in Fig. 7, this gap stems from the ability of larger models to make effective use of backtracking. DeepSeek, for instance, frequently revised its hypotheses when initial assumptions failed, with rationales such as “linear models fail to fit multiple datapoints; explore multiplicative structure." By contrast, Qwen rarely recovered once it committed to an incorrect initial hypothesis, such as assuming linearity when the true equation was nonlinear. This suggests that the capacity to manage and revise multiple hypotheses in context is a key driver of success in larger models under structured reasoning strategies.

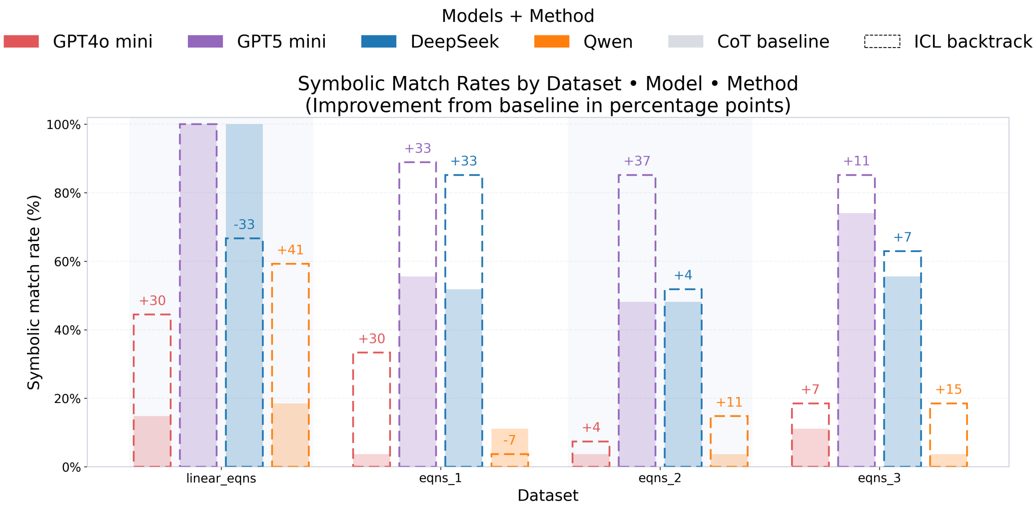

Figure 6: Compares performance of models given the chain-of-thought prompts baseline vs. with ICL backtracking examples. The improvement symbolic match rate is consistent across a majority of the models, with the one exception being DeepSeek for the linear equations dataset. Overall, this figure demonstrates the effectiveness of revision-oriented reasoning traces and structured reasoning prompts over direct prompts.

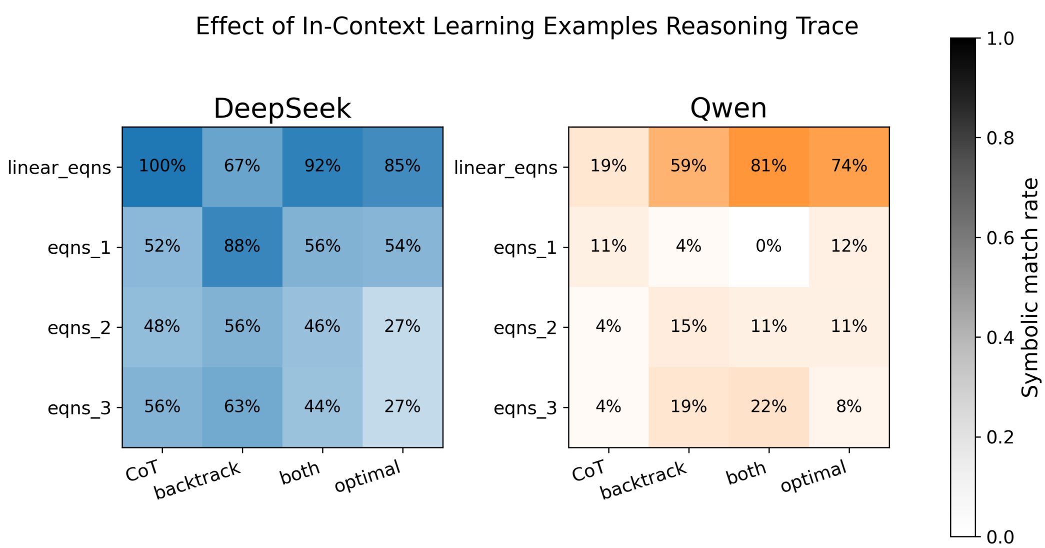

Figure 7: This figure illustrates that larger models make effective use of backtracking as shown by DeepSeek achieving highest symbolic accuracy across all datasets with ICL examples. On the other hand, the smaller model Qwen, achieves lower symbolic accuracy even with ICL examples suggesting that smaller models get stuck within reasoning and are unable to recover from an incorrect initial hypothesis, such as assuming the equation is linear when it is not. In contrast, from our analysis, we found that a common backtrack reason DeepSeek proposes is “Linear models fail to fit multiple datapoints; explore multiplicative structure" suggesting larger models can reasonably hold multiple hypotheses in-context. A general trend we notice is that backtracking also tends to do well for more complex equations.

4.3 Subtask Analysis

To better understand reasoning bottlenecks, we decompose the overall task into two subtasks: (1) cause–effect identification, where the model identifies how changes in individual variables affect the output, and (2) equation formation, where these detected effects must be composed into a symbolic rule as further described in Section 3.

4.3.1 Subtask 1: Cause–Effect Identification

When datapoints are presented in a structured order, all models perform strongly on this subtask, reaching above 80% accuracy (Fig. 8). However, performance collapses when datapoints are permuted such that consecutive data points no longer differ in only one independent variable as they have been shuffled from the datapoint ordering posed in the construction process described in Fig. 3.1. Accuracy drops below 45% across all models when datapoint order is permuted. This sharp decline indicates that models rely heavily on sequential cues to detect cause–effect relationships and are not robust to input order. In other words, while models can pattern match effectively and identify cause-effect relationships between variables from numerical data when the data are aligned, they fail to generalize when the structure is disrupted.

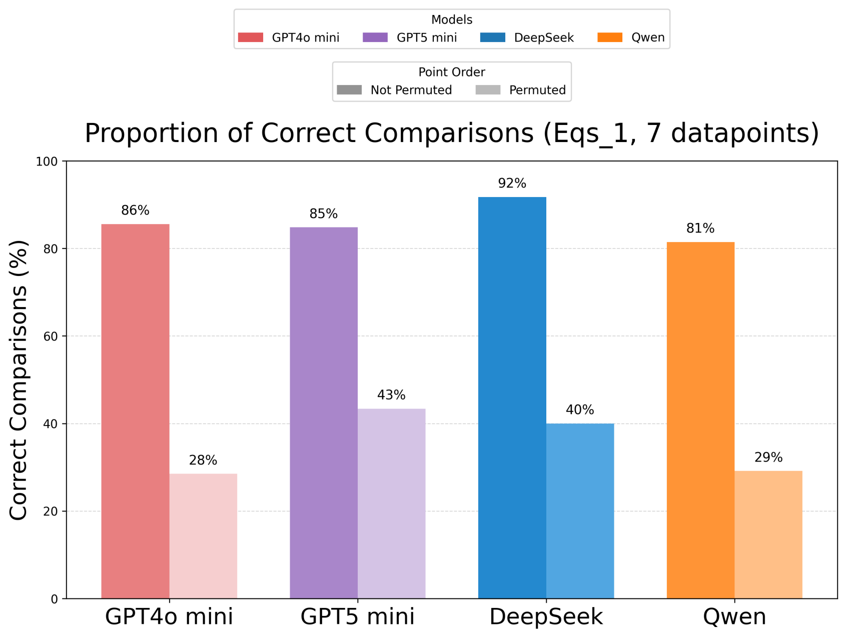

Figure 8: Subtask 1 (Cause-Effect Identification): Accuracy under structured versus permuted datapoint orders. Performance drops from being above 80% for all models to bellow 45% for all the models when inputs are permuted, indicating that when input is structured, models can pattern match effectively, but models are not yet robust to varying input order.

4.3.2 Subtask 2: Mathematical Expression Formation

The second subtask, equation formation (Section 3), emerged as the primary bottleneck for smaller models. Even when correct variable cause-effect information was supplied from subtask 1, smaller models often failed to combine these effects into the correct symbolic rule (Fig. 9). Larger models such as DeepSeek and GPT-5 mini performed substantially better, achieving symbolic match rates of 96% and 81%, respectively. In contrast, smaller models like GPT-4o mini and Qwen rarely exceeded 25%, performing near chance level. The difficulty was especially pronounced on nonlinear equations, where smaller models tended to default to oversimplified linear fits. These results underscore that while LLMs can detect local dependencies reliably, the composition of these dependencies into a full equation remains a challenge for smaller models, limiting their capacity for inductive reasoning in mathematical modeling.

Together, these findings demonstrate that while small and large LLMs can reliably identify cause–effect relationships under structured input, smaller LLMs severely falter at composing these insights into full symbolic expressions. This gap highlights symbolic expression formation as the key obstacle in scaling inductive reasoning for mathematical modeling tasks, especially for smaller models.

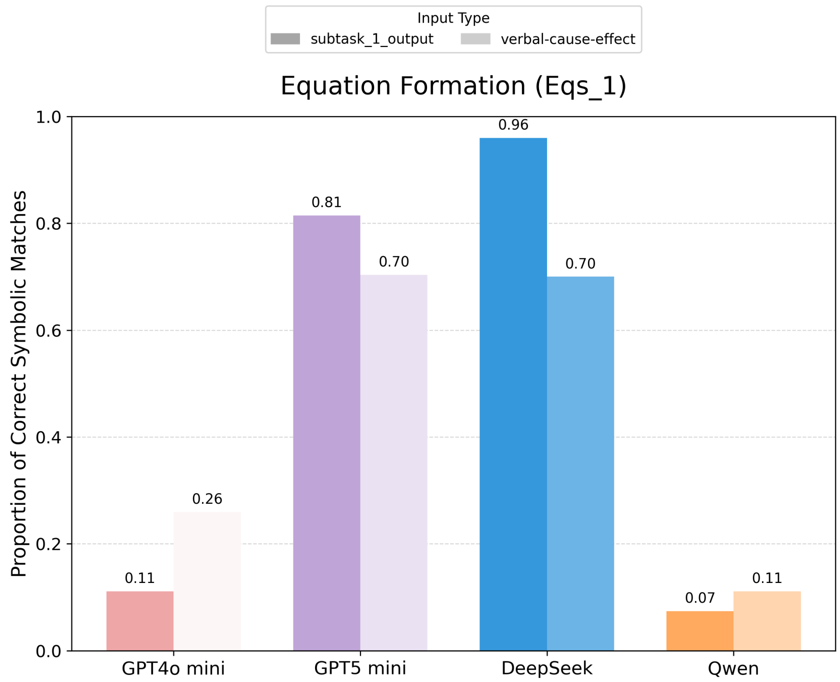

Figure 9: Subtask 2 (Mathematical Expression (i.e. Equation) Formation): Accuracy on equation formation after correct variable–effect identification from subtask 1. Darker bars show performance when using symbolic outputs from subtask 1 as input; lighter bars show performance when given a natural language description of the variable–effect relationships. While larger models (DeepSeek, GPT-5 mini) succeed in forming correct equations with up to 96% accuracy, smaller models (GPT-4o mini, Qwen) remain below 27%. This demonstrates that equation formation is the central bottleneck: models can recognize how individual variables influence the output, but smaller models in particular fail to compose these insights into the correct symbolic form, especially for nonlinear cases.

5 Conclusion

This work introduces a novel benchmark to systematically evaluate LLMs' ability to perform numerical mathematical modeling via inductive reasoning, isolating symbolic rule discovery from linguistic cues. Our results reveal that overall, LLMs can perform interpretable, data-driven reasoning with backtracking and search strategies, which outperform direct and CoT prompts dramatically. However, model size matters, and small models ($<$ 32B parameters) fail without guidance as they often default to linear regression in the first step in over 90% of cases and are unable to formulate nonlinear equations. In contrast, larger models rely on long reasoning traces to identify a nonlinear equation after initially assuming a linear form. Performance and equation complexity are positively correlated. Also, models are often sensitive to order and fail to correctly identify cause-effect relationships from permuted datapoints.

Our results also show that structured reasoning traces—in particular, backtracking and search-based traces—substantially boost symbolic accuracy, especially for nonlinear equations. Larger models benefit most, suggesting an ability to maintain and revise multiple hypotheses in context. Subtask analysis further highlights the gap between relatively easy cause–effect identification and the bottleneck of full equation formation.

Together, these findings underscore both the capabilities and limitations of current LLMs in low-level inductive reasoning. Although LLMs remain constrained by biases and cascading failures, with strategic prompting and reasoning guidance, models can discover interpretable symbolic rules from raw data shedding light into LLMs' broader role in automated modeling and numerical analysis.

6 Future Work

While our benchmark provides a controlled setting to evaluate inductive reasoning, the datasets are synthetic and focus on three-variable equations, which may not capture the full diversity of real-world modeling tasks. Additionally, our evaluation emphasizes symbolic accuracy, but does not yet measure, at scale, the plausibility of approximate fits or partial reasoning paths. We also rely on hand-crafted reasoning traces for in-context learning, which constrains scalability and may not generalize across broader domains and explore other fine-tuning strategies may be appropriate.

This project also connects to ongoing work in the Machine Learning for Education (ML4ED) lab. Prior research has focused on interactive learning environments like PharmaSim, where reinforcement learning (RL) and LLM-based agents adaptively collect and respond to learner data. However, these agents face challenges in identifying which data are most informative for accurate modeling. By uncovering systematic biases in LLMs' inductive reasoning, for example their tendency to default to linear approximations, this work highlights how agents could be guided to collect more relevant and diverse data, thereby improving downstream analysis and feedback. Our benchmark thus serves as a complementary tool to strengthen interactive educational systems by revealing where and how inductive reasoning fails.

Future work can address these limitations in several ways. The benchmark can be expanded to include higher-dimensional datasets. Second, fine-tuning on systematically designed reasoning traces could move beyond few-shot prompting and provide more robust inductive capabilities. Third, probing the internal mechanisms of good inductive reasoning LLMs (such as those that have been fine-tuned to perform well on inductive reasonign tasks) to discover underlying reasoning patterns by analyzing low-level details such as attention patterns, pattern identification, and arithmetic behaviors can help us understand *how* LLMs perform induction, not just whether they succeed. Finally, connecting this line of work to real-world scientific modeling tasks will test whether the gains we see here translate to practical applications in data-driven discovery.

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